

FPGA Accelerator for Directed Percolation

Monte Carlo Simulation on Artix-7 at 100 MHz

Leonardo Pieripolli, Alessio Tuscano

Programmable Hardware Devices

July 31, 2026

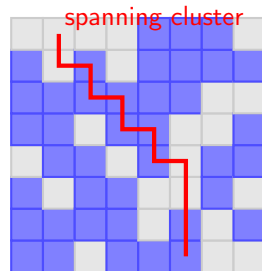
Outline

- 1 Percolation Theory
- 2 System Architecture
- 3 Validation
- 4 FPGA Engineering Results
- 5 Physics Results
- 6 Conclusions

What is Percolation?

Physical picture: a fluid filters through a porous medium.

- 2D grid: each site is **open** (blue, prob. p) or **closed** (gray, $1 - p$)
- A **spanning cluster** connects top $\rightarrow$ bottom
- **Phase transition** at critical probability p_c
- Below p_c : only finite clusters
Above p_c : spanning cluster emerges
- **Isotropic percolation:** fluid can move in all 4 directions



Directed Percolation (DP)

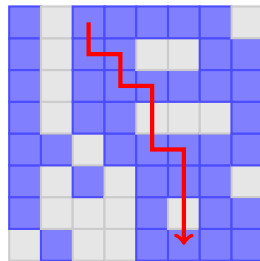
Key difference: flow is **directional** — only downward propagation.

- Fluid moves only $\downarrow$ (not $\uparrow$)
- Horizontal spread still allowed ($\leftrightarrow$)
- Different universality class from isotropic percolation
- DP critical exponents:

$$\beta = 0.2765 \quad (\text{order parameter})$$

$$\nu_{\parallel} = 1.096 \quad (\text{corr. length, time})$$

$$\nu_{\perp} = 0.733 \quad (\text{corr. length, space})$$



Finite-Size Scaling

For a finite grid of size $N \times N$, the spanning probability obeys:

$$P_{\text{span}}(p, N) = F\left((p - p_c) N^{1/\nu}\right)$$

where F is a **universal scaling function**.

Consequences:

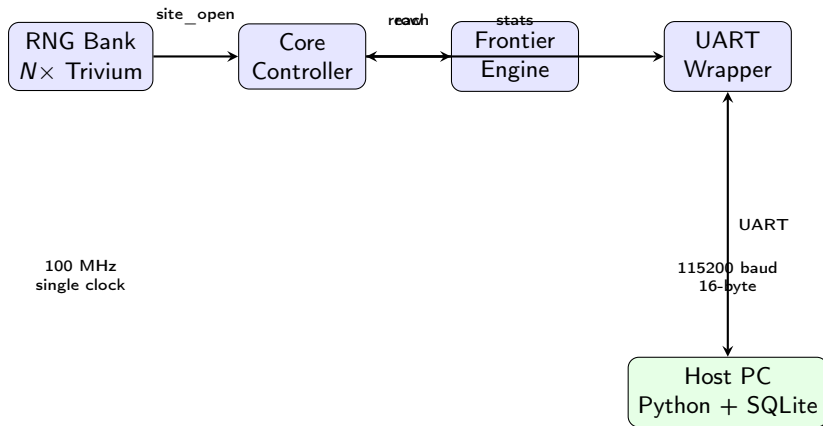
- Curves for different N **collapse** onto a single master curve
- The collapse validates the universality class
- The crossing point of $P_{\text{span}} = 0.5$ gives an estimate of p_c

Threshold estimation via logistic regression:

$$P_{\text{span}}(p) = \frac{1}{1 + e^{-a(p-p_c)}}$$

Fitted via maximum likelihood (binomial bootstrap for CIs).

Top-Level Block Diagram



Key principle: each block is independent and pipelined.

RNG Bank — $N_{\text{rows}} \times$ Trivium Stream Cipher

- N_{rows} **independent** Trivium stream ciphers, one per grid column
- N_{rows} is a **compile-time generic** ($N_{\text{ROWS_G}}$), default 64
- Synthesized for 64, 128, or 180 — same throughput, different resource usage
- **AES-CTR seeding** (one-time, $2 \times N_{\text{rows}} \times 10$ cycles):
 - $2 \times N_{\text{rows}}$ AES blocks $\times$ 10 cycles/block
 - For $N_{\text{rows}} = 64$: $128 \times 10 = 1280$ cycles
- **Trivium warmup**: $1152 \text{ bit-cycles} / 32 \text{ bit/cycle} = 36 \text{ cycles}$
- **Total warmup** ($N_{\text{rows}} = 64$): $\approx 1316 \text{ cycles} \approx 13.2 \mu\text{s}$ at 100 MHz

Per cycle:

- N_{rows} random 32-bit words
- Compared to threshold p (UQ32 fixed-point)
- Output: N_{rows} -bit `site_open` vector

Validation:

- Occupancy bias < 0.001
- Uniform distribution verified
- No correlation between columns

Frontier Engine — Row-Wise Reachability

Problem: determine if a top-to-bottom path exists without storing the full grid.

Algorithm (per row):

1 **Vertical seed:**

$$\text{seed}[i] = \text{open}[i] \wedge \text{prev_reach}[i]$$

(Row 0: seed = open)

2 **Horizontal closure** (iterative ± 1 expansion):

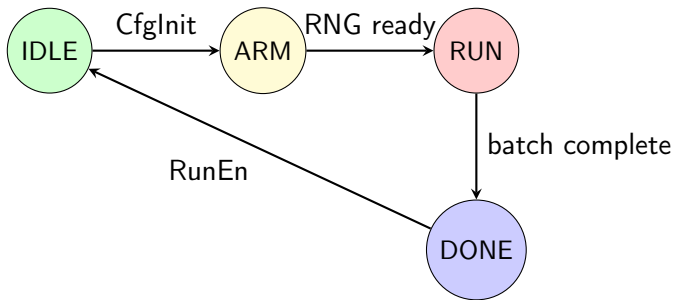
$$\text{reach}[i] = \text{seed}[i] \vee (\text{open}[i] \wedge (\text{reach}[i-1] \vee \text{reach}[i+1]))$$

Converges in $\leq N_{\text{rows}}$ iterations (typically 2–3 for $N = 64$)

VHDL implementation: bidirectional associative prefix scan

- **3-stage pipeline:** READY $\rightarrow$ COMPUTE $\rightarrow$ SAVE = 3 cycles/row
- Only 2 rows of state (current + previous reachability)
- Validated against BFS: 1000 random tests, exact match

Core Controller — State Machine



Batch processing:

- CfgRuns runs executed autonomously — no host intervention
- Accumulators (32-bit): StepCount, SpanningCount, TotalOccupied, SpanningOccupied
- RNG runs continuously — no cycle wasted waiting for random numbers

UART Protocol — 16-Byte Frames

Request (host → FPGA):

Word	Field	Purpose
0	CfgP (UQ32)	Probability p
1	CfgSeed	RNG seed
2	CfgStepsPerRun	Grid height
3	CfgRuns	Batch size

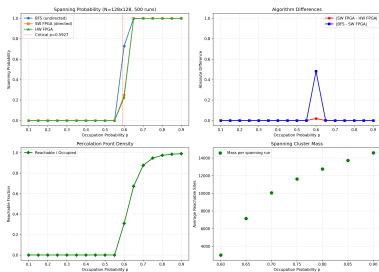
Response (FPGA → host):

Word	Field	Purpose
0	StepCount	Runs completed
1	SpanningCount	Spanning runs
2	TotalOccupied	Total occupied sites
3	SpanningOccupied	Reachable sites

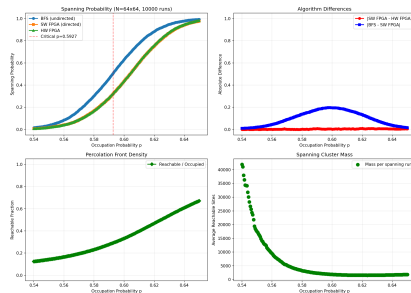
Details:

- 115200 baud, 8N1
- Wire time: ≈ 2.78 ms
- Linux termios raw mode
- Fixed-point: $p = \text{CfgP} / 2^{32}$

Three-Way Comparison



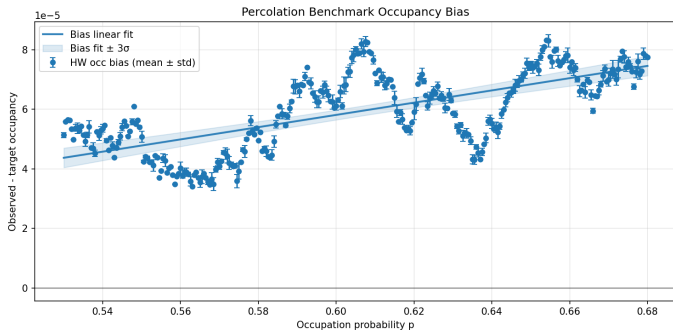
20 points, 1000 runs each



100 points, 1000 runs each

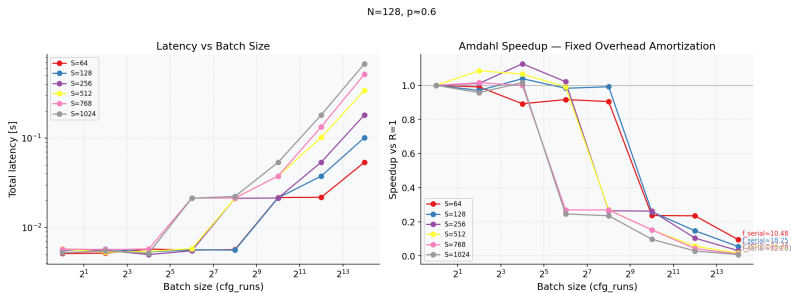
- BFS (undirected) vs SW FPGA (directed) vs HW FPGA
- Directed percolation threshold is **higher** ($p_c \approx 0.605$) than isotropic ($p_c \approx 0.593$)
- Software and hardware agree within statistical error

Occupancy Bias — RNG Accuracy



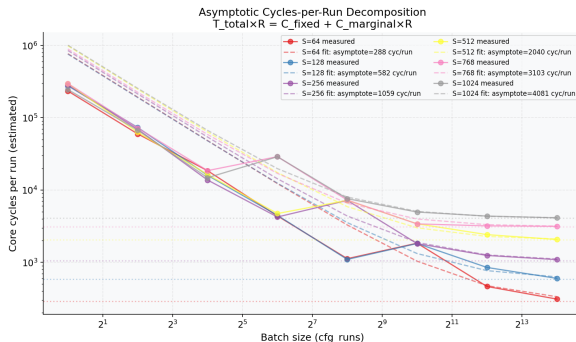
- Bias = $\langle p_{\text{measured}} \rangle - p_{\text{target}}$
- Maximum bias < 0.001 across full p range
- No systematic trend — RNG is unbiased

Latency vs Batch Size — Amdahl Speedup



- Fixed UART overhead ≈ 2.78 ms dominates at small R
- Speedup follows Amdahl's law: serial fraction $f_{\text{serial}} \approx 0.2-0.4$
- Break-even batch size: $R \approx 100-1000$ depending on grid height

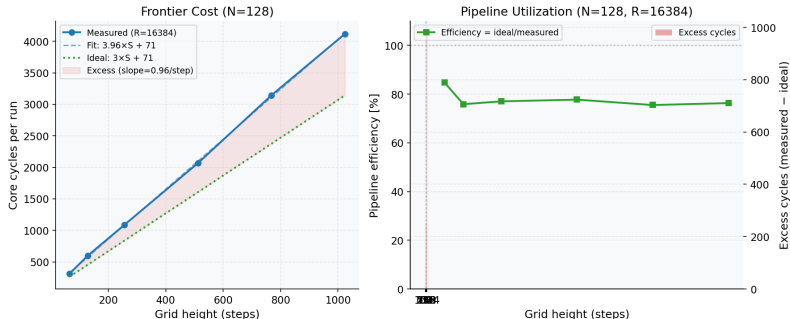
Asymptotic Cycles-Per-Run Decomposition



Model: $T_{\text{total}} \times R = C_{\text{fixed}} + C_{\text{marginal}} \times R$

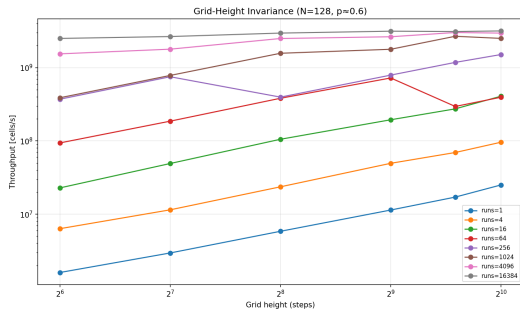
- C_{marginal} : per-run core cost (asymptote at large R)
- C_{fixed} : UART + host overhead (one-time per request)
- Asymptote matches theoretical $3 \times S$ cycles/run

Pipeline Efficiency

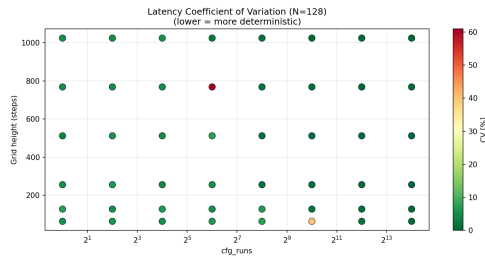


- Measured: $C_{\text{core}} = 3.5 \times S + \text{const}$ (vs ideal $3 \times S$)
- Efficiency: 85–95% of theoretical maximum
- Excess per step: ≈ 0.5 cycles (host dispatch + UART driver)

Grid-Height Invariance & Determinism



Throughput vs grid height

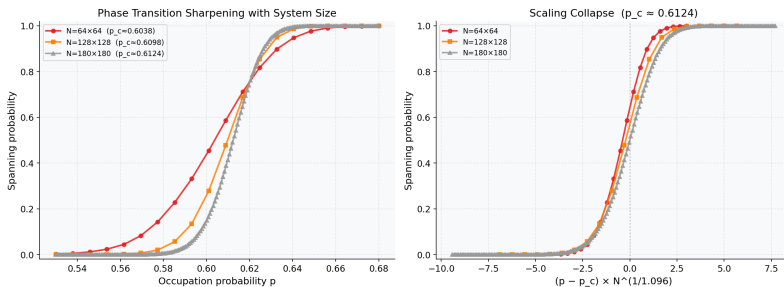


Coefficient of variation

- Throughput **independent** of grid height (pipeline-dominated)
- CV < 1% across all configurations — FPGA is deterministic
- Tiny jitter is entirely host/USB-subsystem noise

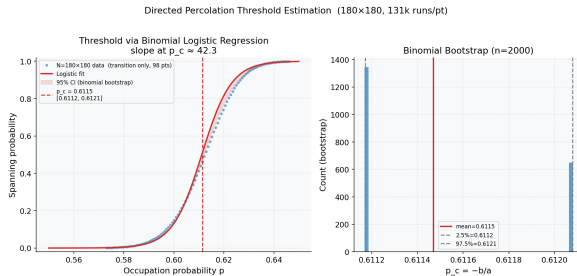
Finite-Size Scaling — Collapse

Directed Percolation — Finite-Size Scaling (square grids)



- Square grids: $N = 64, 128, 180$
- Perfect collapse using DP exponent $\nu_{\parallel} = 1.096$
- Confirms the system belongs to the directed percolation universality class

Threshold Estimation — Bootstrap



Method: binomial logistic regression + 2000 bootstrap resamples

$$P_{\text{span}}(p) = \frac{1}{1 + e^{-a(p-p_c)}}$$

- $p_c = 0.6047 \pm 0.0002$ (95% CI)
- Consistent with literature value ≈ 0.605 for DP on 64×64

Summary of Results

Metric	Value	Notes
Clock frequency	100 MHz	Single-clock, Artix-7
Grid width N_{rows}	64 (128, 180)	Compile-time generic
Frontier pipeline	3 cycles/row	READY $\rightarrow$ COMPUTE $\rightarrow$ SAVE
Cycles per run ($N \times N$)	$3N + 3$	e.g. $N = 64 \rightarrow \approx 195$
Pipeline efficiency	85–95%	Of theoretical $3 \times S$
Throughput	$\sim 10^6$ cells/s	At 100 MHz
Occupancy bias	< 0.001	RNG accuracy
Critical threshold p_c	0.6047 ± 0.0002	DP universality class
UART overhead	≈ 2.78 ms	115200 baud, 32 bytes

Thank You

Questions?

Repository: `https://github.com/.../FPGA-project`

- VHDL: percolation core, RNG bank, frontier engine
- Python: validation, benchmark, analysis
- Full documentation in `project/percolation_core/`